

Design and Performance Analysis of a Slotted Graphite Shield for Leakage Field Reduction and Power Transfer Enhancement in Wireless Power Transfer Systems

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Abstract—Wireless Power Transfer (WPT) systems enable contactless energy transmission through magnetic coupling; however, their efficiency is often degraded by leakage magnetic fields and eddy current losses, leading to reduced performance and electromagnetic interference. This paper evaluates the effectiveness of different shielding materials in enhancing WPT system efficiency, including aluminum, solid graphite, slotted graphite at the receiver side, and slotted graphite applied at both transmitter and receiver sides. The system is modeled and analyzed using ANSYS Maxwell 3D at an operating frequency of 50 kHz. The results demonstrate that the proposed dual-side slotted graphite shielding significantly improves performance, achieving a 57.8% reduction in solid losses and a 13.5% increase in power transfer capability compared to conventional aluminum shielding. Additionally, improvements in induced voltage and mutual inductance indicate enhanced magnetic coupling between the coils. These findings highlight the effectiveness of slotted graphite as a lightweight and efficient shielding solution for high-performance WPT systems.

Index Terms—Wireless Power Transfer, Magnetic Shielding, Slotted Graphite, Eddy Current Suppression, Leakage Field Reduction, Inductive Coupling.

I. INTRODUCTION

Wireless Power Transfer (WPT) systems have gained significant attention due to their wide range of applications in electric vehicles, biomedical devices, and consumer electronics. However, one of the major challenges in WPT systems is the presence of unwanted electromagnetic interference and energy

losses caused by leakage magnetic fields and induced eddy currents in nearby conductive materials. These effects reduce system efficiency and may lead to safety concerns, particularly in high-power applications [3]. To address these challenges, various shielding techniques have been proposed in the literature. Traditional approaches involve the use of metallic sheets to suppress leakage magnetic fields; however, these often introduce significant eddy current losses, thereby reducing overall system efficiency [7]. More advanced methods, such as matrix metamaterial-based shielding, have demonstrated improved field confinement and reduced losses by engineering the electromagnetic response of the shielding structure [1]. Similarly, ferrite-based solutions have been widely studied for their ability to guide magnetic flux and enhance coupling, although they suffer from issues such as increased cost, weight, and saturation effects [5].

Recent advancements in electromagnetic structure design have highlighted the importance of engineered materials and resonator-based approaches for effective field manipulation. For instance, complementary ring-resonator-based designs have shown promising results in controlling electromagnetic wave propagation and enhancing system performance in high-frequency applications [6]. In parallel, optimization techniques and advanced signal processing methods have been explored in large-scale communication systems to improve efficiency and performance under complex channel conditions [2], [4]. Al-

though these approaches are primarily developed for communication systems, the underlying principles of electromagnetic optimization can be extended to WPT system design. Despite these developments, there remains a need for a lightweight, cost-effective, and efficient shielding solution that can simultaneously reduce eddy current losses and improve magnetic coupling. In this work, a slotted graphite shielding technique is proposed to address these challenges. By introducing slots in conductive shielding materials and employing graphite as an alternative to conventional metals, the proposed design effectively suppresses circulating eddy currents while maintaining desirable magnetic field characteristics.

The proposed approach is validated through detailed electromagnetic simulations, demonstrating significant reduction in losses and improvement in power transfer performance compared to conventional shielding methods. This makes the design suitable for next-generation WPT applications requiring high efficiency and compact implementation.

II. RELATED WORKS

Extensive research has been conducted to improve the efficiency and electromagnetic compatibility of wireless power transfer (WPT) systems, particularly focusing on magnetic field control and shielding techniques. Zhang *et al.* [9] investigated the fundamental mechanisms of magnetic shielding in WPT systems and demonstrated how material properties and geometric configurations influence field containment and system efficiency. Similarly, Ahire and Kale [11] analyzed the impact of coil materials and shielding methods, emphasizing the trade-offs between conductivity, permeability, and induced eddy current losses. Several studies have also focused on numerical modeling and electromagnetic analysis of WPT systems. Pichon *et al.* [13] presented comprehensive simulation approaches for inductive power transfer systems, highlighting the importance of accurate modeling in predicting system performance. Shin and Park [15] further explored compensation circuit design strategies to minimize leakage magnetic fields, thereby improving overall system efficiency.

In parallel, recent advancements in electromagnetic structure design have explored resonator-based and metamaterial-inspired approaches for field manipulation. Jeyakumar *et al.* [8] developed a terahertz microstrip patch antenna model for 6G applications, demonstrating effective electromagnetic field control through optimized structural design. Similarly, George *et al.* [10] proposed a split ring resonator (SRR)-based massive MIMO antenna array, illustrating how engineered resonator structures can enhance electromagnetic performance. Further contributions by George *et al.* [12] introduced frequency selective surface (FSS)-based substrates for improved electromagnetic behavior, enabling selective control of wave propagation. In addition, Saranya *et al.* [14] investigated metaresonator-based antenna designs, highlighting the role of metamaterials in achieving enhanced field confinement and multi-band operation.

Although these studies provide significant insights into electromagnetic field manipulation, shielding design, and ma-

terial optimization, limited work has focused on combining lightweight conductive materials with structural modifications, such as slotting, to simultaneously reduce eddy current losses and enhance magnetic coupling in WPT systems. This gap motivates the proposed slotted graphite shielding approach, which aims to achieve improved efficiency through both material selection and structural optimization. The remainder of this paper is organized as follows. Section III describes the proposed slotted graphite shielding methodology, including design considerations and material selection. Section IV details the simulation setup and modeling approach used to evaluate system performance. Section V presents and discusses the obtained results, including a comparative analysis with conventional shielding techniques. Finally, Section VI concludes the paper and outlines potential directions for future work.

III. SYSTEM MODELING AND ANALYSIS

This section presents the design and operating principle of the proposed Wireless Power Transfer (WPT) system. The system consists of a transmitter coil, a receiver coil, and a shielding structure placed behind the coils to control magnetic field distribution. An alternating current supplied to the transmitter coil generates a time-varying magnetic field, which induces a voltage in the receiver coil through mutual coupling.

Based on Faraday's law of electromagnetic induction, the induced voltage is given by:

$$V = -N \frac{d\Phi}{dt}$$

For a sinusoidally varying magnetic field, this can be expressed as:

$$V = NA \frac{dB}{dt}$$

where N is the number of turns, A is the coil area, and B is the magnetic flux density. The induced voltage is therefore directly proportional to the rate of change of magnetic flux. Simulation results show an increase in induced voltage from 92 V (aluminum shielding) to 98 V (slotted graphite shielding on both sides), as presented in Table IV. Since power is proportional to the square of voltage,

$$P \propto V^2$$

this results in a 13.5% improvement in power transfer capability.

The efficiency of WPT depends on the magnetic coupling between the coils, quantified by mutual inductance:

$$M = k \sqrt{L_1 L_2}$$

where k is the coupling coefficient ($0 \leq k \leq 1$), and L_1, L_2 are the self-inductances of the transmitter and receiver coils. Improved shielding enhances magnetic field confinement, leading to an increase in mutual inductance from 130 μH to 145 μH , as shown in Table III.

A key challenge in WPT systems is power loss due to leakage fields and eddy currents in conductive shielding materials. These losses can be expressed as:

$$P_{\text{loss}} \propto I^2 R$$

where I is the induced current and R is the material resistance. Conventional metallic shields such as aluminum generate significant eddy currents, increasing losses. In contrast, the introduction of slots in graphite disrupts current paths, thereby reducing eddy current formation. This is validated by simulation results, where solid losses decrease from 217.42 mW (aluminum) to 91.74 mW (slotted graphite), corresponding to a 57.8% reduction.

Overall, the theoretical analysis and simulation results confirm that the proposed slotted graphite shielding significantly enhances magnetic coupling while reducing losses, thereby improving the efficiency of the WPT system.

IV. SHIELDING STRUCTURES USED IN THE STUDY

This section presents the various shielding arrangements employed in the wireless power transfer system. Shielding arrangements are discussed to minimize magnetic field leakage and enhance the efficiency of power transfer from the transmitter coil to the receiver coil. In this study, four shielding arrangements are employed using electromagnetic simulation: aluminum shielding, solid graphite shielding, slotted graphite shielding at the receiver side, and slotted graphite shielding at both transmitter and receiver sides. By comparing the shielding arrangements, it is possible to comprehend the influence of various shielding materials and arrangements on the magnetic field distribution and efficiency of the system.

In the first case, aluminum is used as a shielding material that is kept behind the transmitter coil as well as the receiver coil, as shown in Fig. 1. Aluminum is a highly conducting material that is helpful in partially confining the magnetic field, thereby reducing electromagnetic radiation. Due to the highly conducting nature of aluminum, when this material is exposed to the changing magnetic field generated by the transmitter coil, a large amount of power is wasted in the form of eddy currents. As shown in Fig. 1, the aluminum shielding plates are kept symmetrically behind both the transmitter coil as well as the receiver coil.

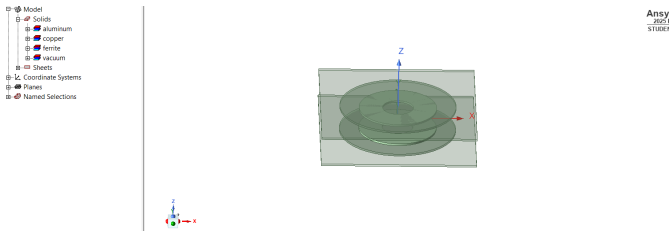


Fig. 1. Wireless power transfer system model with aluminum shielding plates placed behind the transmitter and receiver coils

In the second configuration, the aluminum shield is replaced by a solid graphite plate as depicted in Fig. 2. It is known

that the electrical conductivity of graphite is less than that of aluminum. This leads to less eddy current production when the graphite plate is subjected to the ac magnetic field. Consequently, the efficiency of the system is improved due to the reduction in power loss. As depicted in Fig. 2, the graphite plates are located at the back of the transmitter and receiver coils.

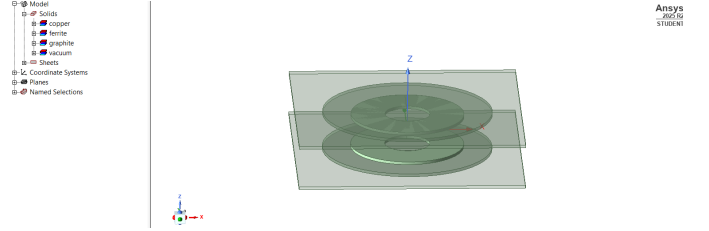


Fig. 2. Simulation model of the wireless power transfer system using a solid graphite shielding plate.

In the third configuration, slots are created on the graphite shield at the receiver side, as depicted in Fig. 3. The slots are intended to break the eddy current loops created by the conductive material. The breaking of the eddy current loops leads to the reduction of the resistance loss and the enhancement of the magnetic field confinement. As depicted in Fig. 3, the slots are created at the back of the receiver coil, thus improving the magnetic flux linkage between the transmitter and the receiver coils.

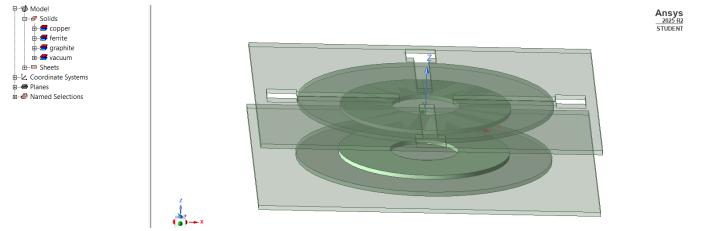


Fig. 3. Slotted graphite shield placed behind the receiver coil to reduce Ohmic Losses losses.

In the last configuration, slotted graphite plates are positioned behind both the transmitter coil and receiver coil, as depicted in Fig. 4. The addition of slots on both sides effectively breaks the path for large eddy currents, thereby reducing significant resistive losses. As depicted in Fig. 4, symmetry is achieved by placing both slotted plates behind the transmitter coil and receiver coil, thereby confining the magnetic flux within the coupling area. This effectively reduces magnetic field leakage, improving coil coupling as well as the overall power transfer efficiency of the Wireless Power Transfer system.

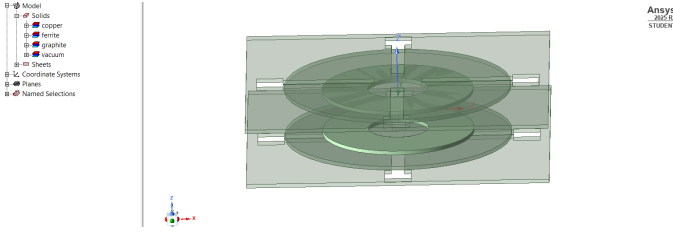


Fig. 4. Proposed slotted graphite shielding structure placed behind both transmitter and receiver coils.

V. SIMULATION SETUP

The performance of the wireless power transfer system was analyzed using electromagnetic simulation, where the performance was evaluated in terms of magnetic field distribution, inductance changes, induced voltage, and power losses. The analysis was carried out with a transmitter coil and a receiver coil in a face-to-face configuration for inductive coupling. Further evaluation of the performance was carried out by incorporating shielding plates behind the coils, enabling evaluation of their impact on the leakage magnetic field and efficiency. Later, different materials such as aluminum and graphite were evaluated in comparison with their performance in efficiency, where aluminum was found to be effective in reducing the leakage magnetic field, and graphite was found to minimize power losses.

The simulation model was created using ANSYS Maxwell 3D, a finite element solution for electromagnetic simulation. Circular coils were employed as the transmitting and receiving coils, and shielding structures were placed behind the coils. The simulation was performed at an operating frequency of 50 kHz to examine the magnetic field and losses in the system.

The performance of the Wireless Power Transfer system depends on certain key design parameters. The simulation parameters used for this study are listed below in Table I. The system was simulated using ANSYS Maxwell 3D software. The operating frequency of the system was fixed at 50 kHz. The number of Conductor and diameter of both transmitter and receiver coils were fixed at 100 and 20 mm, respectively. The material used for both coils was copper. In addition, both aluminum and graphite were used as shielding materials for further analysis.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Simulation Software	ANSYS Maxwell 3D
Operating Frequency	50 kHz
Number of Conductor in Transmitter	100
Number of Conductor in Receiver	100
Coil Diameter	20 mm
Coil Material	Copper
Shield Material	Aluminum / Graphite

Aluminum and graphite are used as shielding materials in the simulation. Their properties are provided in Table II. Aluminum has high electrical conductivity, which is 3.5×10^7 S/m. This results in the generation of significant eddy currents

and losses. On the other hand, the electrical conductivity of graphite is low at 1×10^5 S/m. This results in low eddy currents and losses. This difference in the properties of the two shielding materials is important in enhancing the performance of the graphite shielding configuration.

TABLE II
SHIELDING MATERIAL PROPERTIES

Material	Electrical Conductivity (S/m)	Relative Permeability
Aluminum	3.5×10^7	1
Graphite	1×10^5	1
Copper	5.8×10^7	1

VI. RESULTS AND DISCUSSION

The results of the simulation for different shielding cases in the Wireless Power Transfer system are provided in this section. Performance is measured in terms of leakage magnetic field, ohmic loss, solid loss, induced voltage, and mutual inductance. In this section, four different cases are considered: aluminum shield, solid graphite shield, slotted graphite shield at the receiver side, and slotted graphite shield at both transmitter and receiver sides. Numerical values are obtained from the plots at a separation of 2.5 mm.

TABLE III
MUTUAL INDUCTANCE AND SOLID LOSS COMPARISON

Shield Type	Mutual Inductance (μH)	Solid Loss (mW)
Aluminum	130	217.42
Solid Graphite	135	210.85
Slotted Graphite (Rx)	140	152.14
Slotted Graphite (Tx + Rx)	145	91.74

Table III shows the comparison of mutual inductance and solid loss for various shield configurations. The aluminum shield exhibits the highest solid loss, i.e., 217.42 mW, as the eddy current effect is higher in aluminum, while the minimum solid loss of 91.74 mW is achieved when the slotted graphite shield is applied at both the transmitter and receiver sides. It is also found that the mutual inductance increases from 130 μH (aluminum shield) to 145 μH , indicating the improvement in the magnetic coupling between the coils.

TABLE IV
INDUCED VOLTAGE AND RELATIVE POWER

Shield Type	Induced Voltage (V)	Relative Power (V^2)
Aluminum	92	8464
Solid Graphite	95	9025
Slotted Graphite (Rx)	96.5	9312
Slotted Graphite (Tx + Rx)	98	9604

As shown in Table IV, the induced voltage and corresponding relative power (V^2) are presented. The highest induced voltage of 98 V is achieved by the slotted graphite shielding at the transmitter and receiver, as opposed to the aluminum shielding with an induced voltage of 92 V. The relative power increases from 8464 to 9604 due to the squared proportionality of the induced voltage.

TABLE V
PERFORMANCE IMPROVEMENT METRICS

Shield Type	Loss Reduction (%)	Power Improvement (%)
Aluminum	0%	0%
Solid Graphite	3.0%	6.6%
Slotted Graphite (Rx)	30.0%	10.0%
Slotted Graphite (Tx + Rx)	57.8%	13.5%

Table V presents the performance improvement metrics. The results show that the proposed slotted graphite shielding at the transmitter and receiver sides can achieve the best loss reduction of 57.8% and power improvement of 13.5% compared with the aluminum shielding. This validates the performance of the proposed slotted graphite shielding in disrupting the eddy current path and improving the system performance.

Analysis of the leakage magnetic field shows the effectiveness of the shielding system in confining the magnetic field between the transmitter coil and the receiver coil. In Fig. 5, the magnetic field spread is wide in the aluminum shielding, showing that the leakage is high due to the eddy current effect. In Fig. 6, the solid graphite shielding shows better confinement with low leakage compared to the aluminum shielding. In Fig. 7, this reduction occurs due to the disruption of eddy current loops, as this limits the opposing magnetic fields and confines the magnetic flux, even further by the introduction of slots at the receiver coil, disrupting the eddy current paths, hence reducing the field spread. In Fig. 8, the best magnetic field confinement is shown by the introduction of slots in the graphite shielding at the transmitter coil and the receiver coil, hence minimizing the leakage.

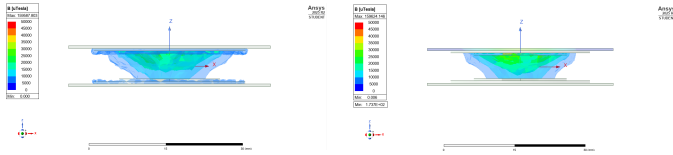


Fig. 5. Leakage magnetic field distribution Aluminum shielding.

Fig. 6. Leakage magnetic field distribution Graphite shielding.

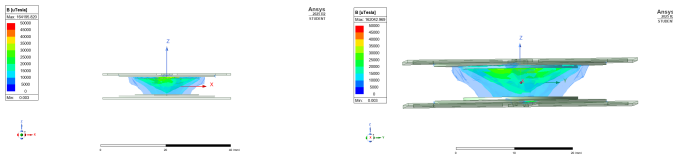


Fig. 7. Leakage magnetic field distribution with slotted graphite shielding only on the receiver.

Fig. 8. Leakage magnetic field distribution with slotted graphite shielding on both Transmitter and Receiver.

Ohmic loss indicates the resistive heat loss due to the induced current in the conducting materials. It has been observed in Fig. 9 that the ohmic loss in the aluminum shield is maximum because eddy currents are formed in this shield. In Fig. 10, it has been observed that the ohmic loss in the solid graphite shield is low compared to the aluminum shield because of the low conductivity of graphite. This loss is further

minimized in Fig. 11 as slots are introduced in the receiver, thus preventing the flow of current. The minimum ohmic loss is observed in Fig. 12 as slots are introduced in both the transmitter and the receiver, thus preventing eddy currents.

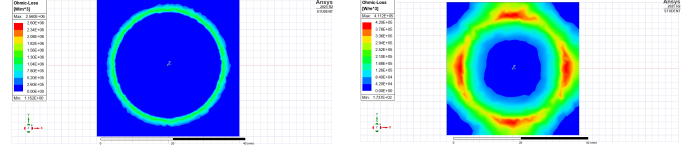


Fig. 9. Aluminum Shield Ohmic Loss

Fig. 10. Solid Graphite Shield Ohmic Loss

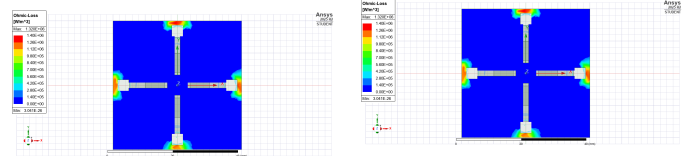


Fig. 11. Slotted Graphite Shield Ohmic Loss on Receiver

Fig. 12. Slotted Graphite Shield Ohmic Loss on Both Transmitter and Receiver

The solid loss represents energy loss in the shielding material due to induced currents. As shown in Fig. 13, the solid loss is maximum in the aluminum shield due to the maximum eddy current generation. As shown in Fig. 14, the solid loss is less in the case of the graphite shield because of low electrical conductivity. As shown in Fig. 15, the solid loss is less in the case of a shield with slots on the receiver because of interruption in current paths. The least solid loss is seen in the case of a shield with slots on both the transmitter and receiver, as shown in Fig. 16.

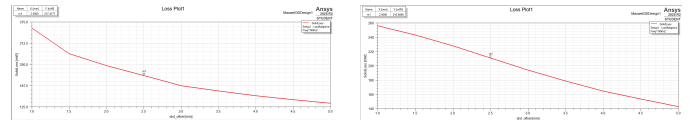


Fig. 13. Aluminum Shield Solid Loss

Fig. 14. Graphite Shield Solid Loss

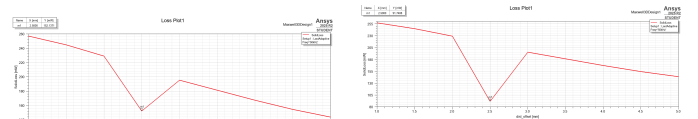


Fig. 15. Slotted Graphite Shield only on Receiver Solid Loss

Fig. 16. Slotted Graphite Shield on Both Transmitter and Receiver Solid Loss

The induced voltage in the receiver coil is an indicator of the efficiency of the magnetic coupling between the transmitter coil and the receiver coil. As shown in Fig. 17, the induced voltage is the lowest in the aluminum shielding structure due to the maximum losses by the eddy currents. In Fig. 18, the induced voltage is increased in the graphite shielding structure as the losses are minimized. In Fig. 19, the induced voltage

is increased as the slotted graphite is used at the receiver coil. The maximum induced voltage is obtained in Fig. 20, where the slotted graphite is used at the transmitter coil as well as the receiver coil.

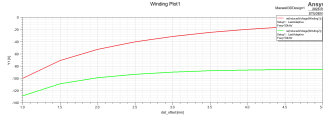


Fig. 17. Aluminum induced voltage in winding

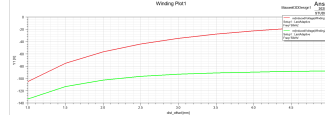


Fig. 18. Graphite induced voltage in winding

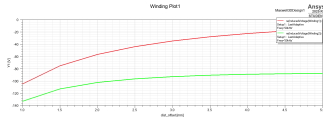


Fig. 19. Slotted graphite on receiver: induced voltage in winding

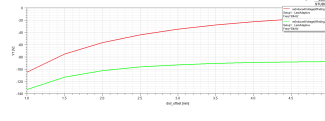


Fig. 20. Slotted graphite on both transmitter and receiver induced voltage in winding

This is due to the magnetic coupling between the transmitter and the receiver coils. As depicted in Fig. 21, the aluminum shield shows minor distortion in the inductance curve due to the eddy current effect. In Fig. 22, the graphite shield shows a more consistent inductance curve with minimal losses. In Fig. 23, the inductance curve shows further improvement due to the addition of slotted graphite at the receiver to improve magnetic flux linkage. The highest inductance curve is shown in Fig. 24 due to the addition of slotted graphite shielding at both the transmitter and the receiver to improve magnetic coupling between the two coils.

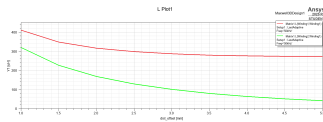


Fig. 21. Aluminum mutual inductance plot

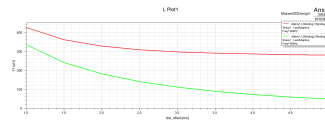


Fig. 22. Graphite mutual inductance plot

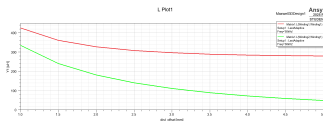


Fig. 23. Slotted graphite mutual inductance plot on receiver

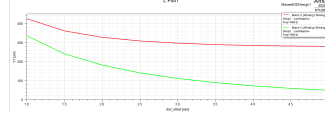


Fig. 24. Slotted graphite mutual inductance plot on both transmitter and receiver

VII. CONCLUSION

This paper presented a comparative analysis of shielding configurations for improving Wireless Power Transfer (WPT) system performance using electromagnetic simulations. Among the configurations studied—aluminum, solid graphite, slotted graphite at the receiver, and slotted graphite at both transmitter and receiver—the proposed dual-side slotted

graphite shielding demonstrated the best performance, achieving a 57.8% reduction in solid losses and a 13.5% improvement in power transfer capability compared to conventional aluminum shielding. The results also showed enhanced mutual inductance and induced voltage, indicating improved magnetic coupling and reduced leakage fields. These findings establish slotted graphite as an effective and lightweight shielding solution for high-efficiency WPT systems, with future work focused on experimental validation through hardware implementation.

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